

Automatic dental biofilm detection based on deep learning

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Abstract

Aim: To estimate the automated biofilm detection capacity of the U-Net neural network on tooth images.

Materials and Methods: Two datasets of intra-oral photographs taken in the frontal and lateral views of permanent and deciduous dentitions were employed. The first dataset consisted of 96 photographs taken before and after applying a disclosing agent and was used to validate the domain's expert biofilm annotation (intra-class correlation coefficient = .93). The second dataset comprised 480 photos, with or without orthodontic appliances, and without disclosing agents, and was used to train the neural network to segment the biofilm. Dental biofilm labelled by the dentist (without disclosing agents) was considered the ground truth. Segmentation performance was measured using accuracy, F1 score, sensitivity, and specificity.

Results: The U-Net model achieved an accuracy of 91.8%, F1 score of 60.6%, specificity of 94.4%, and sensitivity of 67.2%. The accuracy was higher in the presence of orthodontic appliances (92.6%).

Conclusions: Visually segmenting dental biofilm employing a U-Net is feasible and can assist professionals and patients in identifying dental biofilm, thus improving oral hygiene and health.

KEYWORDS

artificial intelligence, dental biofilms, neural networks, photograph, preventive dentistry

Clinical Relevance

Scientific rationale for study: Dental biofilm identification is complex and time consuming. Disclosing agents are commonly used for plaque visualization but have multiple disadvantages. Therefore, automatic dental biofilm detection would benefit both professionals and patients.

Principal findings: A deep learning method showed adequate performance for detecting dental biofilm on intra-oral photographs of permanent and deciduous dentitions with or without orthodontic appliances.

Practical implications: Automated dental biofilm identification in intra-oral photographs may assist oral healthcare providers and laypersons in self-care in the future. An automatic dental biofilm detection tool can be developed based on deep learning.

1 | INTRODUCTION

Dental biofilm is composed of microorganisms and is the primary aetiological factor of caries and periodontal diseases, which are the most prevalent oral diseases globally (Marsh, 2010; GBD 2017, 2018). The mechanical control of dental biofilm through toothbrushing and interdental biofilm removal is an effective strategy for preventing these diseases (Chapple et al., 2015; Jepsen et al., 2017). However, it is difficult for professionals and the population to visualize the biofilm because of its amorphous and low-contrast characteristics, which may impede its efficient removal. Within this context, an easy-to-use tool that assists in dental biofilm detection would positively impact oral health.

Disclosing agents have been used to improve dental biofilm visualization. These agents change the colour of the dental biofilm by retaining dye substances, allowing its observation. Nevertheless, these agents have many drawbacks, such as staining composite restorations, having an objectionable taste, and causing allergic reactions (Lang et al., 1972; Shrivastava et al., 2021). Fluorescent disclosing agents have been used to overcome some disadvantages; however, they require a special light source to make the biofilm visible (Lang et al., 1972). Therefore, an automatic dental biofilm identification tool that does not require disclosing agents would be especially advantageous. artificial intelligence (AI) techniques may support the development of such tools, as they have shown promising results for detection and diagnosis purposes (Schwendicke et al., 2020).

AI is a large field that aims to develop machines that mimic human cognitive functions (Norvig & Russell, 2021). One of the subfields of AI is machine learning (ML), where computers learn from the data using intelligent algorithms to perform a task. ML can potentially exploit meaningful relationships within a dataset, which can be used to diagnose, treat, and predict outcomes of many clinical scenarios (Amisha et al., 2019). Deep learning (DL) uses a complex structure of algorithms that enables the processing of data such as documents, images, and text. In summary, DL is a specialized subset of ML, which is a subset of AI. A convolutional neural network (CNN) is a DL algorithm that can take in an input image, assigning importance (learnable weights) to various aspects/objects in the image while differentiating one from another.

CNN works by receiving an image, employing several trainable filters that weight the pixels, and then categorizing the input image or parts of it. Most studies in healthcare and dentistry employ ML techniques based on supervised learning (LeCun et al., 2015). Supervised learning requires labelled data for training. A domain expert usually does the labelling in a process called annotation (Lin et al., 2014; Liu et al., 2019). In the training stage of supervised learning, the DL model produces an output that is compared with the correct label to adjust the trained algorithms.

CNNs have been used for tooth segmentation and numbering in panoramic radiographs by our group (Jader et al., 2018; Silva et al., 2018, 2020; Pinheiro et al., 2021). Regarding dental biofilm, three studies have evaluated the identification of dental biofilm on digital photographs. The first was able to classify the biofilm level on the images by using a quantitative light-induced fluorescence camera (Imangaliyev et al., 2017). In the second study, a CNN predicted dental biofilm location on digital images of a small dataset (Yaune

et al., 2017). The third study used intra-oral cameras and a biofilm-disclosing agent, showing no significant difference between the DL model and the human specialist on biofilm identification (You et al., 2020). These studies did not have dataset variability, which is required for enhanced network performance in a broader range of scenarios (Renard et al., 2020). In fact, previous studies did not specify sample characteristics (Imangaliyev et al., 2017; Yaune et al., 2017), but only classified the patient's level of dental biofilm or used only photographs of primary teeth with biofilm-disclosing agents, and had findings restricted to intra-oral camera images (You et al., 2020).

The present study used a varied dataset composed of intra-oral photographs of children and adults without and with different orthodontic appliances (metallic and aesthetic braces, fixed expander). Furthermore, U-Net, a type of CNN, not previously used in the context of dental biofilm, was employed for object segmentation, which is the process of classifying each pixel of an input image as biofilm or non-biofilm. This study aimed to estimate the automated biofilm detection capacity of a U-Net neural network on permanent and deciduous tooth images without biofilm-disclosing agents.

2 | MATERIALS AND METHODS

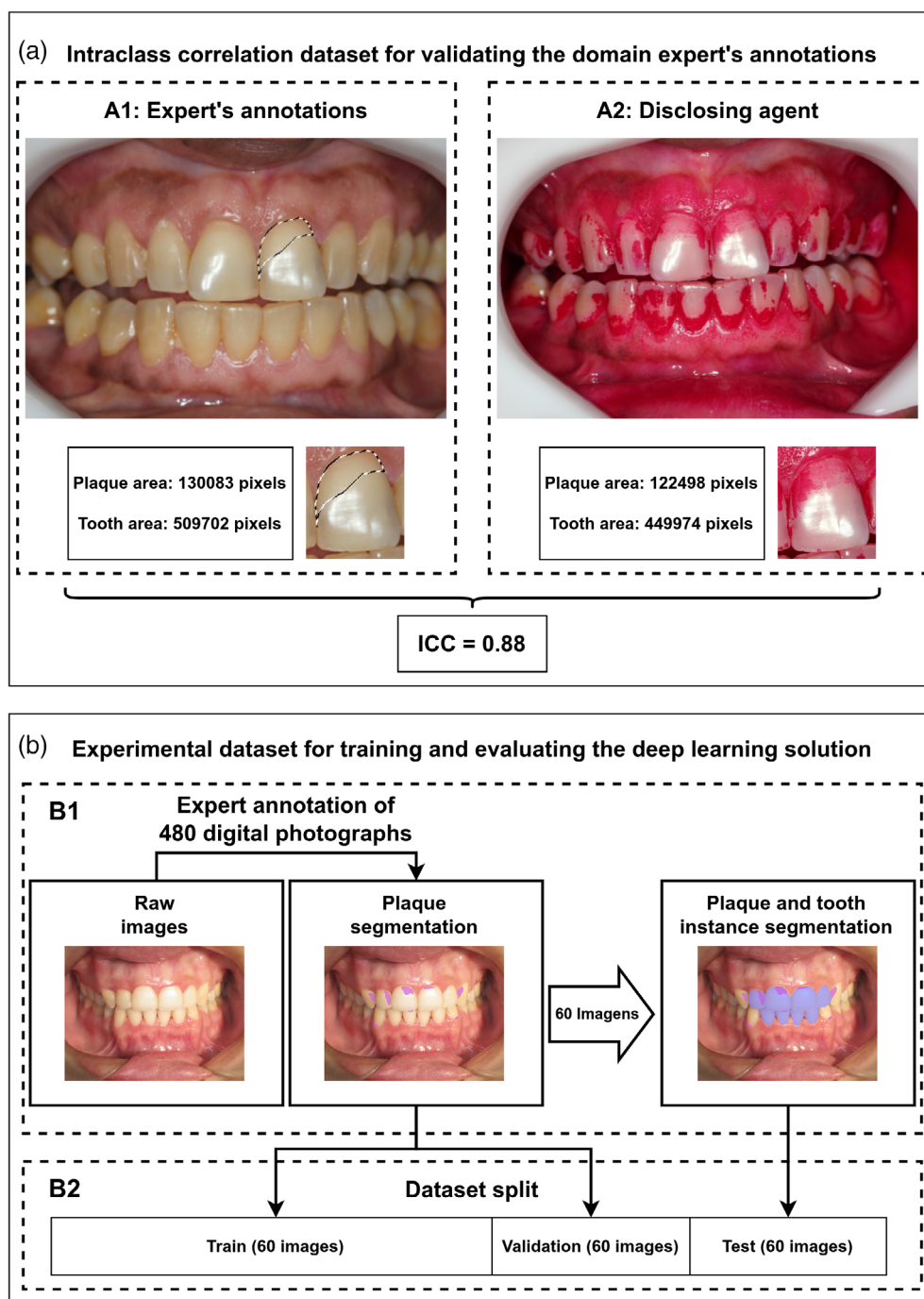
The study was approved by the Ethics Committee of the School of Dentistry, Federal University of Bahia (Protocol number: 4.434.730) and was conducted in accordance with the Declaration of Helsinki. This study was described according to the Checklist for AI in Medical Imaging (Mongan et al., 2020).

2.1 | Study design and data collection

For this retrospective feasibility study, two datasets were collected from an image bank of a private office. The images were randomly selected between 10 September 2019 and 6 March 2021. The first dataset (Figure 1a), the intra-class correlation one, consisted of 96 photographs from 16 patients taken before and after applying a disclosing agent (Tablets Eviplac; BiodinamicasTM) and was used to validate the examiner's biofilm annotation. The second dataset (Figure 1b), the experimental one, comprised 480 intra-oral photos from 160 patients, with or without orthodontic appliances, and without disclosing agents, and was used to train and evaluate the U-Net network to segment the biofilm. The sample size of the experimental dataset was empirically determined based on prior experience (Jader et al., 2018; Silva et al., 2018, 2020; Pinheiro et al., 2021). The eligibility criteria were as follows: No clinically visible caries or tetracycline stain, and frontal and lateral (left and right) photographs taken in the perpendicular plane in the buccal views. Images showing poor light, saliva bubbles, or out-of-focus areas were excluded from this study. Half the patients had dental appliances. All the images were identified by assigning randomized numeric identifiers to each subject.

All photos were taken with a plastic lip retractor using a Canon EOS REBEL T3 camera (4272 × 2848 pixels; approximately

FIGURE 1 Annotations of the datasets. (a) Intra-class correlation coefficient (ICC) dataset: an intra-oral photograph illustrating the procedure for comparing the examiner's annotation (a1) and the disclosing agent outcome (a2). (b) Experimental dataset. (b1) intra-oral photograph illustrating the manual annotation of biofilm segmentation and tooth instance segmentation, and (b2) splitting the dataset into the training, validation, and test sets.



3 megabytes), a Canon Macro 100 lens, and a Canon ring flash MR-14, and were stored in JPEG format. The camera settings were: f/29 aperture; ISO 200; shutter speed 1/160; and auto white balance. The flash was set in the E TTL mode.

2.2 | Consistency of examiner's annotation

One dentist with 29 years of clinical experience annotated the images. Areas of yellowish colour and/or brightness loss and

rough or granular appearance were considered dental biofilm (Figure 1a1). The examiner's annotation was validated by using the intra-class correlation dataset by comparing the dental biofilm area in each tooth of the digital images with and without the disclosing agent. Dental biofilm area was labelled using the Adobe Photoshop software CS5 and its lace tool for automatic area computation (Figure 1a2). The intra-class correlation coefficient (ICC) was 0.93 (95% CI, 0.92–0.94). Therefore, dental biofilm labelled by the dentist (without disclosing agents) was considered the ground truth.

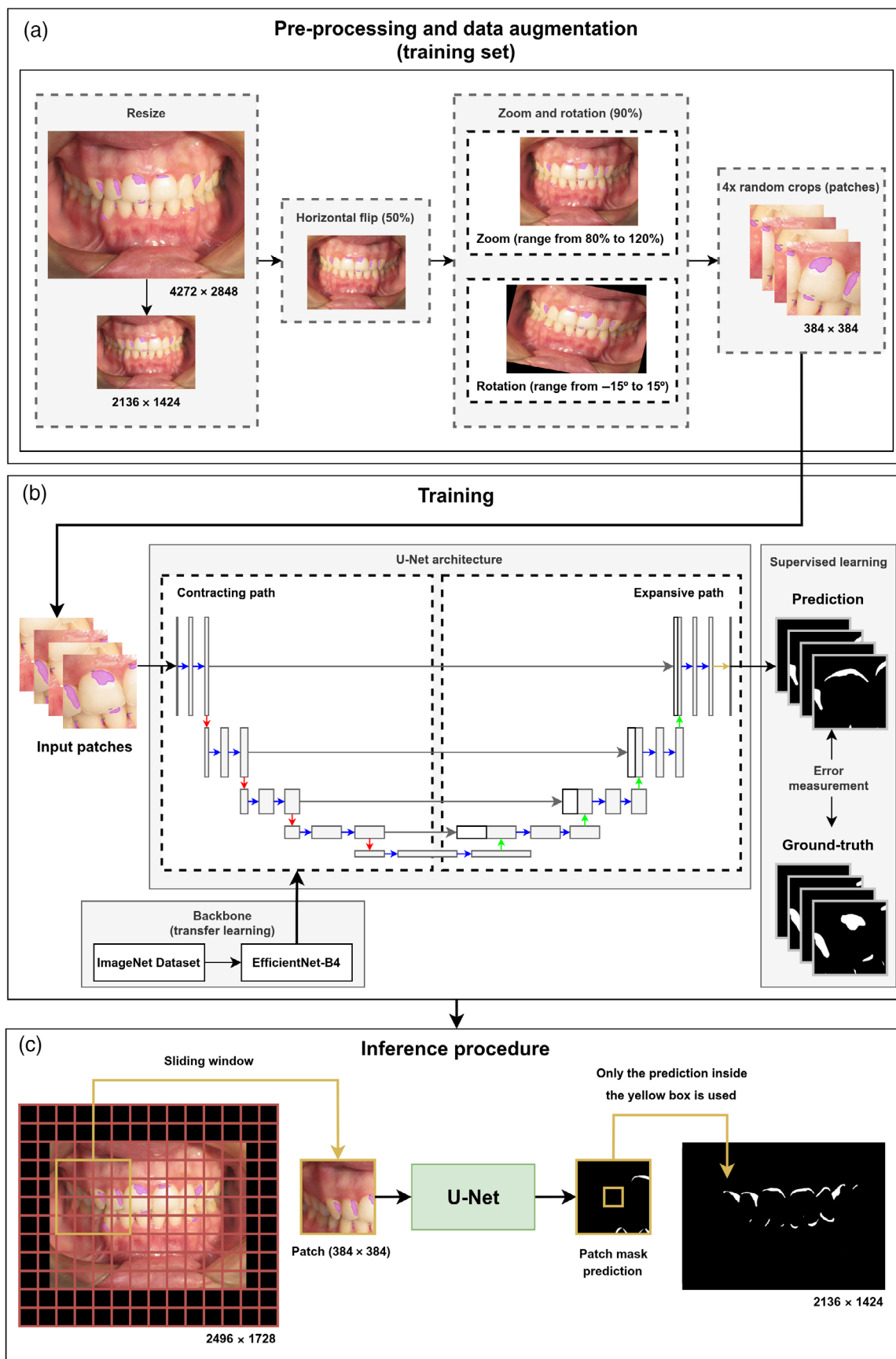


FIGURE 2 Legend on next page.

2.3 | Annotation of the experimental dataset

Two types of annotations were created for the experimental dataset using the polygon tool of the open source image segmentation software COCO Annotator version 0.11.1. The first type was dental biofilm labelling, which provided binary masks for pixel-wise separation of biofilm from the background. These binary masks were used in the training and evaluation procedures. The second type was the tooth instance segmentation, which was used only during the assessment of the neural network and for statistical analysis (Figure 1b1). The buccal surface of each tooth was set as a region of interest. The annotations were performed using frontal view photographs to segment dental biofilm of the incisors and lateral view photographs for the cuspids, premolars, and first molars.

2.4 | Data partitions

The dataset was then randomly split into three subsets: training (360 images), validation (60 images), and test (60 images) (Figure 1b3). To avoid data snooping, different images were included in each subset, and each patient's photographs were included in the same subset.

2.5 | Model

A Red/Green/Blue (RGB) image was taken as input. Rather than inputting the complete high-resolution images to the model, the 384×384 patches were used. Patches sizes were chosen based on a series of preliminary tests, which showed an optimal result with an EfficientNet-B4 trained with a patch size of 384×384 compared with 768 and 1152 patches and full image (Figure S1). A U-Net architecture (Ronneberger et al., 2015) was applied to image segmentation (Figure 2), employing a publicly available code (Yakubovskiy, 2019). A network named EfficientNet-B4 (B4 corresponds to the version of the EfficientNet) (Tan & Le, 2019) was chosen as the backbone for the U-Net architecture. EfficientNet-B4 architecture was chosen because it showed the best results in our preliminary experiments (Figure S2), in which other versions of the EfficientNet network were evaluated, such as B0–B3 and B5–B7.

2.6 | Training

All images were pre-processed by decreasing their widths and heights by 50%, resulting in a final dimension of 2136×1424 .

During training, the data were artificially augmented. Figure 2a illustrates the data augmentation procedure: (i) horizontal flipping 50% of the time, (ii) 80%–120% random zoom, and (iii) -15° to 15° random rotation 90% of the time. After every augmentation step, four random 384×384 patches from each image were selected for training the U-Net in a supervised learning fashion (Figure 2b). The training procedure was conducted without time concerns with 8 Tesla V100 16GB GPUs hardware through a supervised learning approach. The validation procedure (Figure 2c) was carried out after analysing all images, during training. Prediction scores of centre of the patches were used. The prediction score ranged from 0 to 1, with 0 corresponding to the lowest confidence level and 1 to the highest level, because a sigmoid function was used (Christakopoulou et al., 2016). For dichotomous categorization, detection of dental biofilm was defined when the prediction scores outputted by the network exceeded the midpoint, 0.5 (Esposito et al., 2021). The model that obtained the highest F1 score in the validation dataset was kept.

2.7 | Assessment of the U-Net model

Two types of analysis were conducted to evaluate whether the proposed DL solution could correctly segment biofilm: quantitative and qualitative. First, quantitative analysis was performed using the dichotomous categorization (described in the Training Section) in which the model-detected dental biofilm was compared with the ground-truth masks labelled by the dentist. Subsequently, qualitative analysis was performed on the model's worst and best quantitative results per patient, comparing the ground-truth binary masks to the dichotomous predictions and the output prediction scores.

2.8 | Statistical analysis

The annotations by the dentist were compared with the DL model results in a pixel-wise fashion, resulting in true-positive, true-negative, false-positive, and false-negative counts for each pixel. The following metrics were computed per tooth: accuracy, sensitivity, specificity, and F1 score, also known as the Dice similarity coefficient. The accuracy was calculated as the ratio between the number of correct predictions and the total number of predictions. Linear regression was used to assess the association between the model performance and the presence of dental biofilm. All statistics were computed using Python and the SciPy package version 3.7.7.

FIGURE 2 Pipeline of the pre-processing, data augmentation, training, and evaluation procedures. (a) Pre-processing and data augmentation. The training dataset's photographs and their labels were re-sized and data augmentations were applied at each step. At the end of each step, four random patches were selected. (b) Training. The training procedure employed supervised learning, in which the model's outputs were compared with the ground truth to adjust the convolutional neural network. Transfer learning was employed through an EfficientNet B4. (c) The model's inference procedure used a sliding window approach employing only the prediction's centre of each patch.

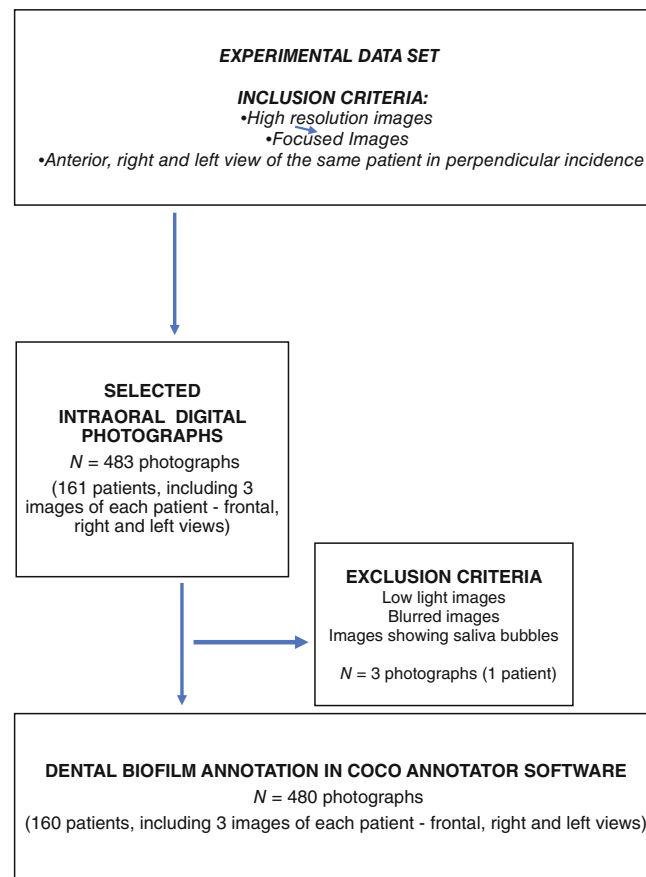


FIGURE 3 Flow of cases, indicating inclusion and exclusion.

Image type	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 score (%)
No orthodontic appliance	91.0	70.9	93.1	59.7
Orthodontic appliance	92.6	63.7	95.6	61.5
Frontal view	91.9	66.9	94.2	57.9
Lateral view	91.8	67.4	94.5	61.7
Total	91.8	67.2	94.4	60.6

TABLE 1 Accuracy, sensitivity, specificity, and F1 score for the model performance in the entire test dataset and subgroups: with and without orthodontic appliances or frontal and lateral views.

3 | RESULTS

Figure 3 shows the flow of cases, indicating inclusion and exclusion. Patients aged 5–73 years had all teeth in the mouth and had no dental caries or periodontal diseases.

3.1 | Quantitative analysis

Table 1 shows the average metrics according to the presence of orthodontic appliances and image type (frontal or lateral view). For the U-net model, the mean accuracy was 91.8% and the mean specificity was 94.4%. The mean F1 score was 60.6% and was higher in the presence of orthodontic appliances (61.5%). The mean sensitivity was

67.2%. Linear regression revealed a positive association between the amount of biofilm per tooth type and the prediction of biofilm by the model. The coefficient of determination R^2 revealed that the linear regression model explained 53.3% of the F1 score, according to biofilm fraction per tooth type as depicted in the scatter plot (Figure 4).

3.2 | Qualitative analysis

The worst results were generally obtained with images that contained a lower amount of dental biofilm. The model segmented the biofilm in the cervical third and inter-proximal regions of teeth showing a lower dental biofilm fraction. In contrast, the best results were obtained with images with a large amount of dental biofilm, even in cases with

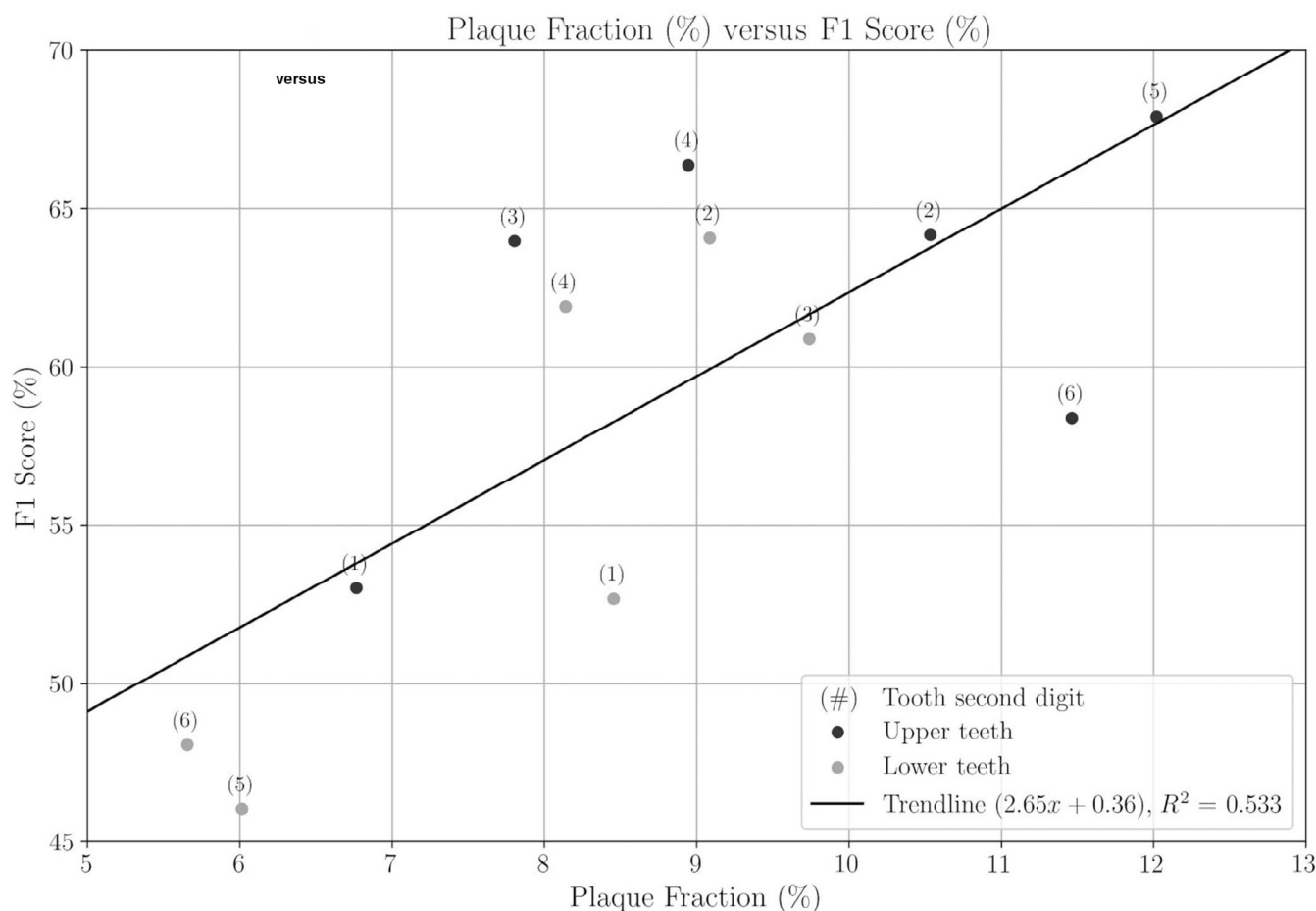


FIGURE 4 Scatter plot and trendline showing the positive linear relationship between F1 score and dental biofilm fraction per tooth type.

orthodontic appliances. Segmentation samples of the model's best and worst performance per patient are depicted in Figure 5.

4 | DISCUSSION

This study evaluated a U-Net network, a DL method, for segmenting dental biofilm on intra-oral photographs of permanent and deciduous dentitions with or without orthodontic appliances. The model's performance was acceptable, with a mean accuracy of 91.8%, F1 score of 60.6%, specificity of 94.4%, and sensitivity of 67.2%, demonstrating that the model was adequate for dental biofilm segmentation.

The resulting model achieved high accuracy and specificity but a moderate F1 score and sensitivity. The moderate F1 score and sensitivity were associated with the complexity of labelling dental biofilm on photographs, as it shows undefined and amorphous borders. This difficulty was confirmed by the values of intra-examiner F1 score (66.3%) on the test set (data not shown) and by the F1 score (66.9%) observed on the validation set (Table S1). The similarity of these F1 scores also indicated no overfitting in this study. A previous study obtained an F1 score of 76% using quantitative light-induced fluorescence images (Imangaliyev et al., 2017). Another study reported an

accuracy of 84.67% (Yauney et al., 2017). Nevertheless, the dataset comprised only permanent teeth. A previous work (You et al., 2020) on automatic biofilm measurements used a small dataset in which each image included only one deciduous tooth. The dataset of the present study was diverse, including deciduous and permanent teeth with and without orthodontic appliances. These features make the training of the model and the achievement of adequate performance more challenging. However, this dataset is more representative of the diversity observed in the population and, therefore, more suitable for generalization. Furthermore, the image evaluation was performed on first molars, premolars, canines, and incisors instead of cropped teeth. This is an advantage since the traditional intra-oral imaging of dental records is performed using this approach, allowing for the use of this model for available photographs. An intra-oral photograph that comprises only one tooth would require high resolution to be cropped through imaging software. A camera capable of capturing high-quality images or a macro lens that could achieve this approximation would be necessary (Renard et al., 2020).

The model showed better performance, that is, a higher F1 score and accuracy, when a more considerable amount of biofilm was observed, following what is expected to be seen with the naked eye in clinical practice. In other words, dental biofilm detection by the

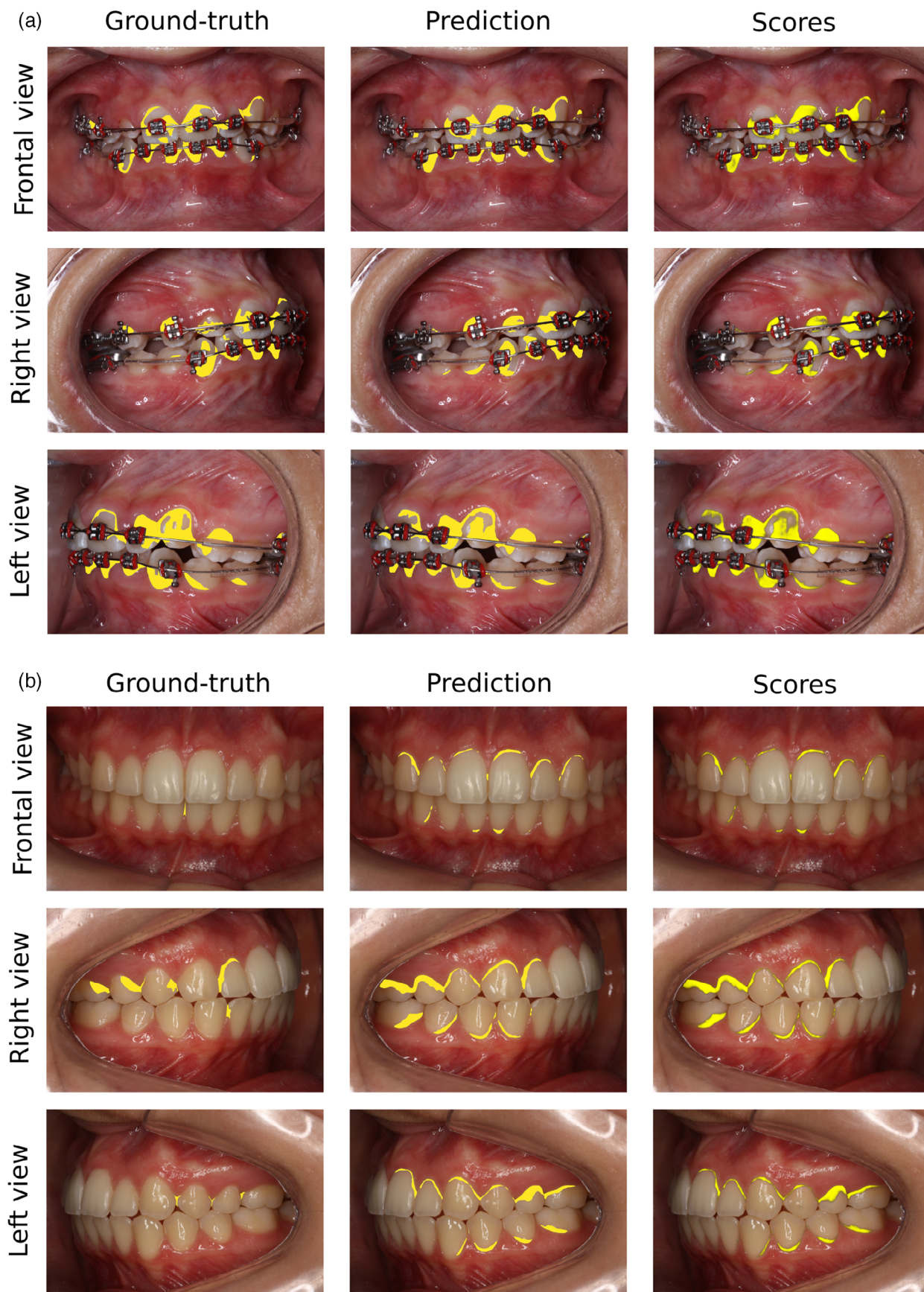


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model and the naked eye is easier when there is more biofilm accumulation. The regression model confirmed this relationship, revealing that a 1% increase in biofilm fraction implied a 2.7% increase in the F1 score (Figure 4). It is worth mentioning that the regression model was used to show the relationship between the amount of dental biofilm and the model result, and not to show that the amount of biofilm completely explains the model performance.

Quantitative and qualitative analysis revealed that the performance of the model was higher in canines, premolars, and first molars, and in the cases with orthodontic appliances, situations in which the removal of dental biofilm by the patients are more complex, and dental biofilm accumulation is more significant (Ren et al., 2014). The analysis also showed that the worst results were obtained for photographs with a lower biofilm fraction. In these cases, the model indicated the presence of biofilm in the cervical third and inter-proximal regions of the teeth, which are prone to biofilm accumulation. In clinical practice, this model behaviour would be valuable to indicate the need for better oral hygiene. The tooth colour did not interfere with the U-Net performance (automatic analysis performed on teeth without orthodontic appliance, Figure S3).

Manual labelling of the images, a step required for the supervised learning techniques, was the greatest challenge of this work, as the visualization of dental biofilm on photographs is a complex and time-consuming task. The annotations were created using the polygon tool of the COCO Annotator software, which allowed fewer interactions and more accurate segmentation than the brush tool since the operator could simply click on the edges of the dental biofilm area and did not have to paint the entire area. A total of 420 images for training and validation were labelled with an average of 240 h of annotation work. This number can be considered a small volume of data by DL standards (Wataya et al., 2020). Then, transfer learning technique, in which the weights of a neural network previously trained on a large dataset are partially tuned or fine-tuned over a smaller dataset, and data augmentation were employed to address this issue. These techniques enhance the dataset, allowing the model to achieve a greater classification accuracy (Shin et al., 2016; Shijie et al., 2017).

In the present study, a more suitable neural network architecture for dental biofilm segmentation, the U-Net, was used, while previous studies classified biofilm levels employing VGG16 or used less suitable networks (ResNet-50 and DeepLabv3+) for biofilm segmentation (Imangaliyev et al., 2017; Yauney et al., 2017; You et al., 2020). U-Net was employed in the present methodology because it is considered the most suitable and feasible architecture for biomedical segmentation, including regions with undefined borders and few annotated images (Ronneberger et al., 2015; Liu et al., 2020). In this study, the U-Net encoder, the EfficientNet-B4, was pre-trained on the ImageNet dataset (Ronneberger et al., 2015; Liu et al., 2020). Additionally,

patches with the exact size of the input images of the pre-trained encoder were used, which optimized the transfer learning procedure because it preserved the image proportions perceived during pre-training and avoided excessive memory consumption required for high-resolution images. Reducing the photograph resolutions would be ineffective because image details are required for biofilm segmentation. It is worth noting that the metrics used in the present study were able to evaluate the model's performance and are the most frequently used. Many other metrics, such as validation loss and area under the receiver operating characteristic curve, are available (Hicks et al., 2022).

Among the limitations of this study, the generalization of the present model is restricted since all photographs were taken with the same camera and under standardized conditions, showing high quality, which is difficult to achieve in practice. Although the imaging protocol for intra-oral photographs is largely standardized, there are variations in image acquisitions, such as equipment type, image resolution, focus, working distance and lighting, angle, and position of the photograph. Model generalization can be expected in images showing similar standardization to the present study. This factor limits the applicability of the current tool by the general population. Furthermore, only one dentist labelled the data. Preferably, more dentists should label the images to avoid rater bias. However, the reproducibility of the dental biofilm evaluation performed by the present dentist was satisfactory (ICC = 0.93), and the concordance with a second examiner was also high (Data not shown: inter-examiner ICC of 0.88, 95% CI 0.85–0.91). Another limitation was that the second and the third molars were not included in the present evaluation because these teeth frequently could not be well visualized in the photographs.

Developing a tool for automatic dental biofilm detection would benefit both professionals and patients since the identification of dental biofilm may be difficult and time consuming because of its amorphous characteristic and colour similarity to the tooth, in either photographs or mouth. Although running a probe along all teeth or using dental biofilm-disclosing agents facilitates biofilm detection, automating a process optimizes manual tasks. The present results show that the CNN model can be used to segment dental biofilm on intra-oral digital photographs and that it is feasible to develop a tool for automatic dental biofilm identification. A protocol that could be applied in different clinical situations to replace the traditional biofilm identification would require taking standardized intra-oral radiographs, if they are not yet available, downloading them to a computer, and running the model through the images. Automatic biofilm detection and quantification could be used as a diagnosis, monitoring, and motivational tool in clinical and research settings.

Future studies may consider adapting the neural network for application to cell phone images, making dental biofilm identification

FIGURE 5 Mosaics of the best (a) and worst (b) performances of the model on the test dataset. The first column shows the ground truth computed through the expert manual annotation, the second is the model prediction, and the third is the model prediction scores for dental biofilm presence. In the last column, the yellow degree of transparency represents the prediction scores: the more transparent, the closer to 1; and the more opaque, the closer to 0.

available on a large scale and thus contributing to adequate dental hygiene and consequent improvement of the population's oral health. An additional advantage of using CNNs for biofilm segmentation is providing a prediction score that indicates teeth and areas where the patient should improve dental biofilm removal, improving the quality of self-performed biofilm control. In future research, adding a clinical examination as a ground truth and including more diverse clinical scenarios, such as the presence of fixed tooth, implant-supported restorations, staining, different degrees of teeth crowding, and caries lesions, would further increase the external validity of the tool.

5 | CONCLUSIONS

The DL CNN model was able to automatically segment visually detectable dental biofilm on intra-oral photographs automatically. The potential for significant model improvement exists as more labelled training data become available. The automated classification of intra-oral photographs may assist oral healthcare providers and laypersons in self-care.

AUTHOR CONTRIBUTIONS

Katia Montanha Andrade and Bernardo Peters Menezes Silva contributed to the preparation of the manuscript. Luciano Rebouças de Oliveira and Patricia Ramos Cury supervised the work in their respective research fields.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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